

Microtopography explains variations in seedling community structure in Panama's tropical moist forests

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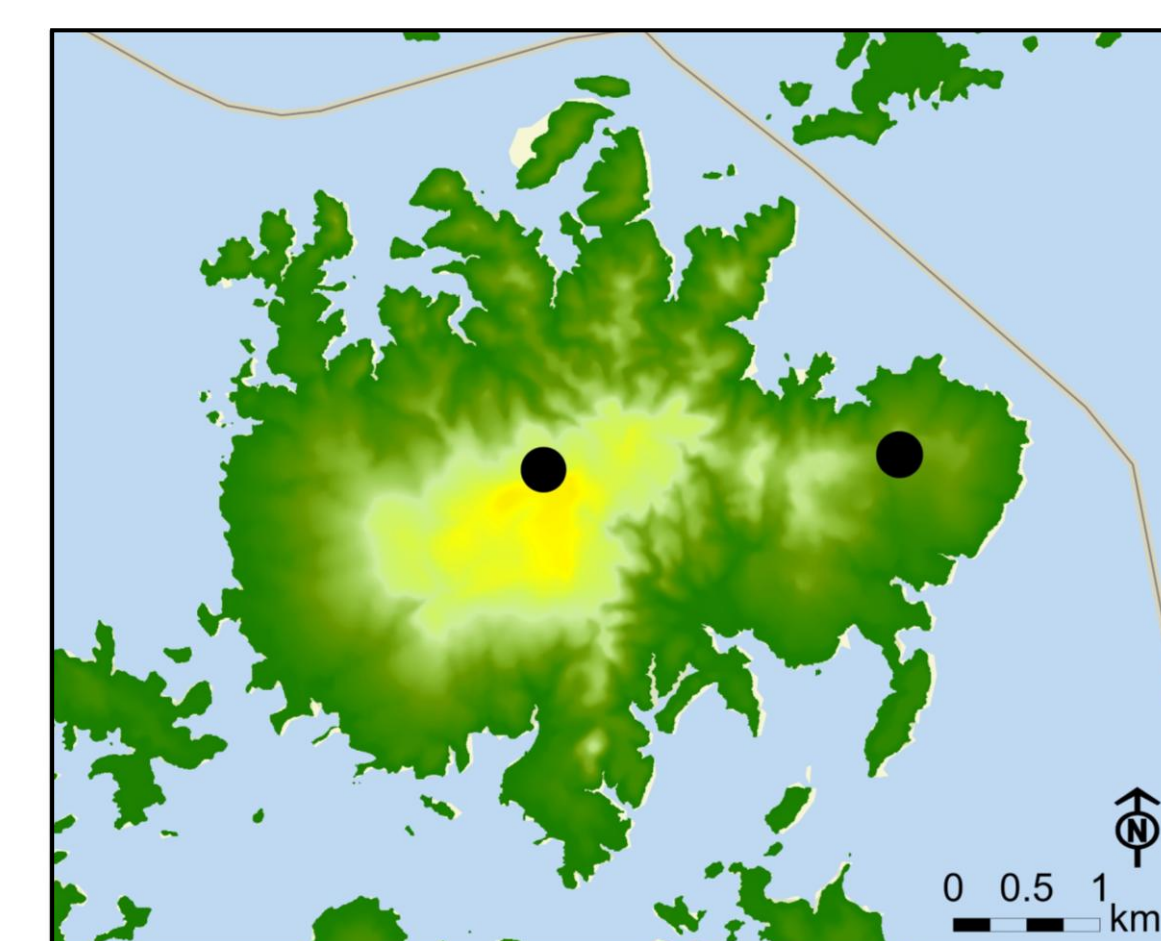
Introduction

Topography has been shown to influence seedling dynamics in tropical forests by forming microhabitats that shape environmental filtering (Daws et al., 2005; Inman-Narahari et al., 2014) and modulate species' functional responses to the density of neighboring vegetation (O'Brien et al., 2022).

Previous work on topographic drivers of seedling performance has primarily focused on topographic habitat categories, such as ridges, valleys, or plateaus. In this study, we aim to explore the relationship between topography and seedling dynamics along continuous axes, and at finer scales, in order to isolate and more precisely quantify how specific topographic conditions may account for fine-scale variations in seedling community structure and function.

Methods

We examined 2 1-hectare late successional (129-year-old) forest stands on Barro Colorado Island, Republic of Panama. Each stand contained 400 1mx1m plots situated in a uniform rectangular grid, with plot centers distributed in 5-meter increments.



Map 1. Locations of 1-hectare late successional forest stands on Barro Colorado Island.

We used linear mixed models to determine the individual and combined effects of topographic variables on plot-level total percent seedling cover and species richness, with stand as a random effect. Model selection was performed using an exhaustive search of all predictor combinations, including interaction terms, and then comparing AIC scores among converged models.

We measured total percent seedling cover as the percentage of a 1mx1m plot comprised of seedling individuals < 1 meter in height.

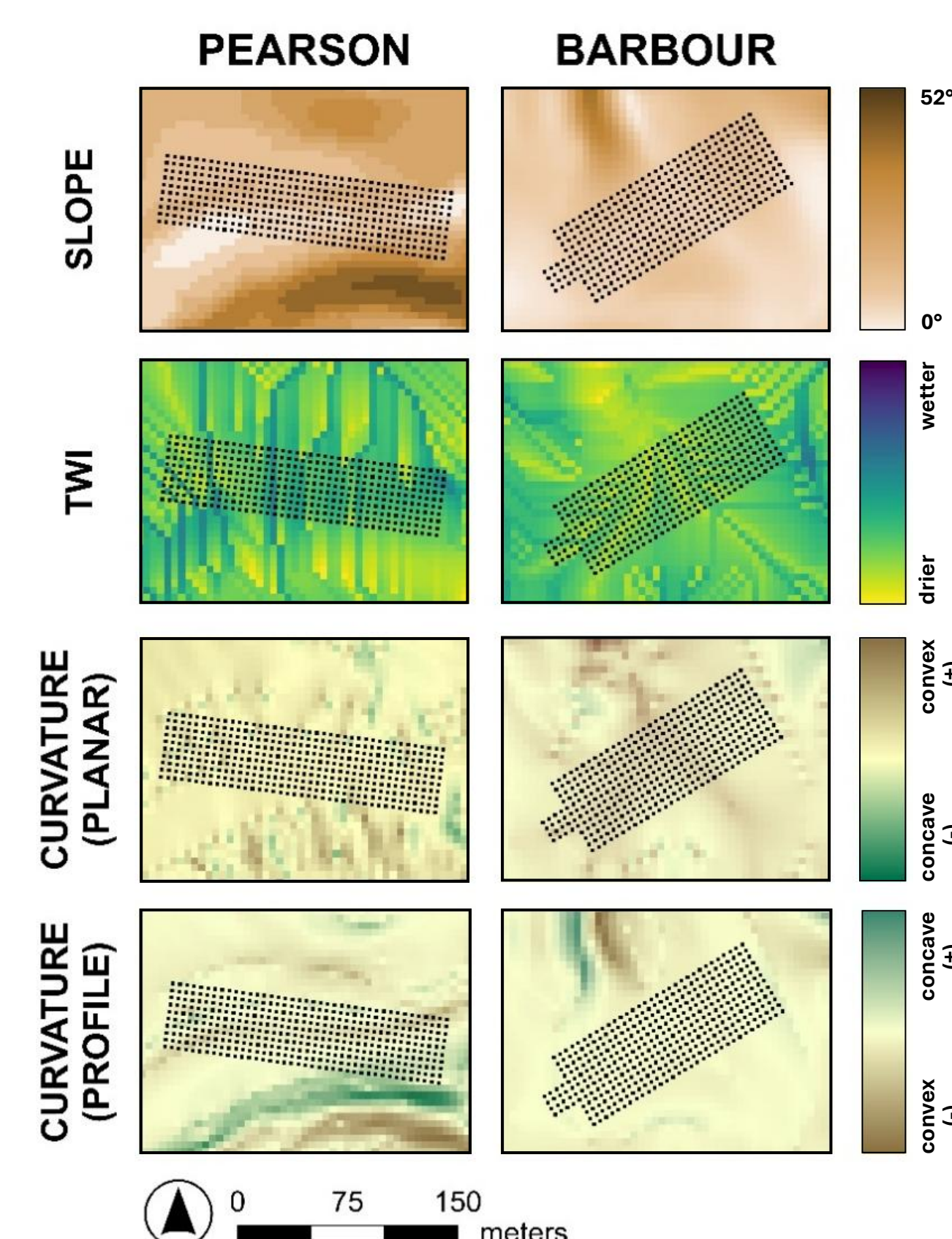


Fig. 1. We examined topography via 8 variables, each derived from a 5mx5m resolution digital elevation model (DEM). Variables which exhibited significant associations with seedling response variables are displayed.

TWI, or topographic wetness index, is a measure of soil moisture, or an area's proclivity to inundation.

Results

Linear mixed effects models showed that a plot's combination of slope and planar curvature explained significant variations in total percent seedling cover ($p < .05$) (Fig. 2).

More steeply sloped plots exhibited lower total percent seedling cover ($p < .05$).

However, this relationship was complicated by coinciding planar curvature. When slope increased with planar convexity, plots saw higher cover. Concavity, however magnified slope's negative relationship with cover ($p < .01$).

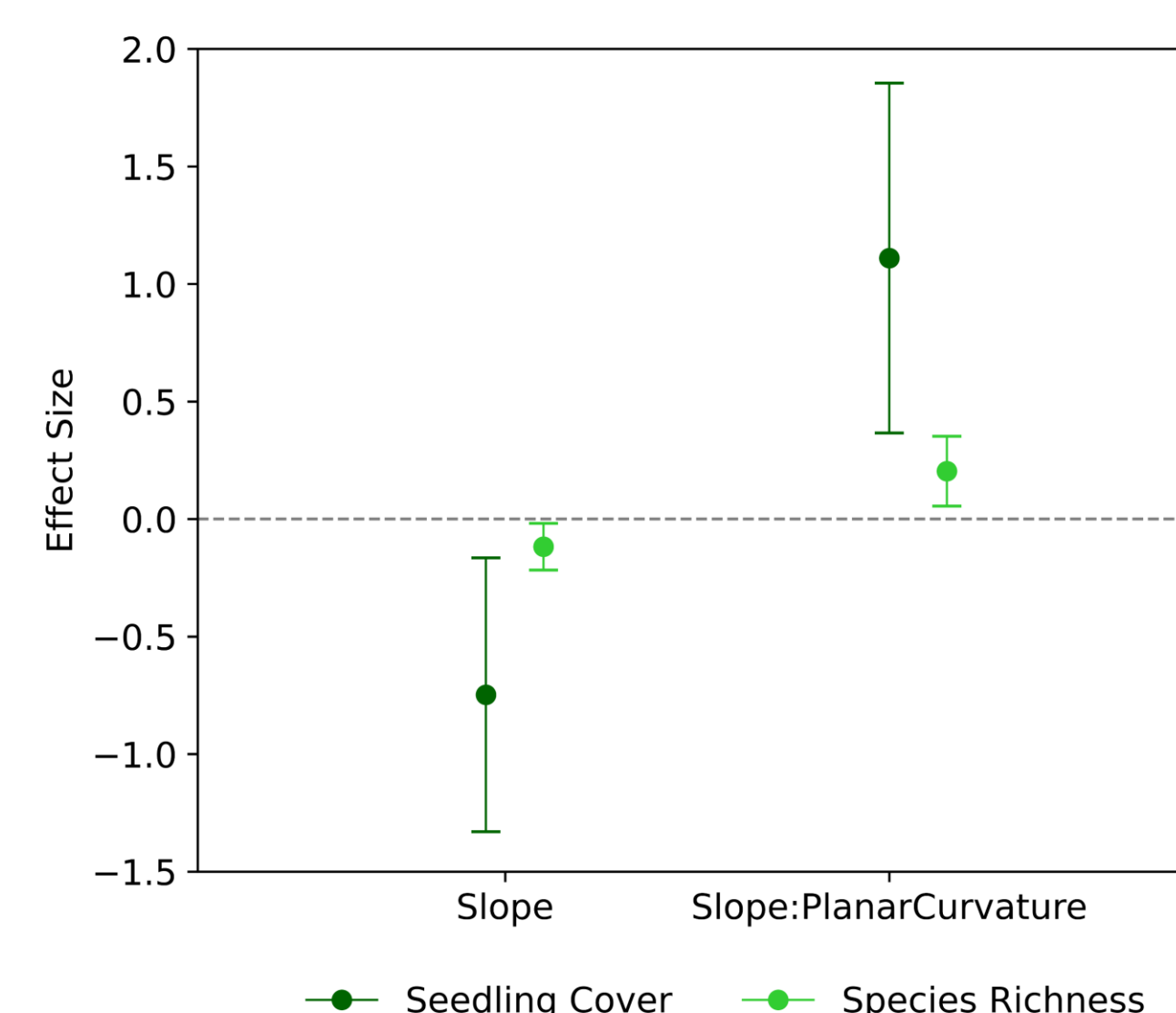


Fig. 3. Planar curvature alters slope's relationship with cover significantly more than its relationship with species richness.

More concave plots (i.e., micro-valleys) exhibited higher species richness ($p < .01$) when topographic wetness was unaccounted for. However, in the presence of increasing topographic wetness, species richness became positively associated with profile convexity ($p < .01$).

This hints that species richness may exhibit differential relationships to micro-valleys and micro-ridges depending on those features' positions within the broader watershed.

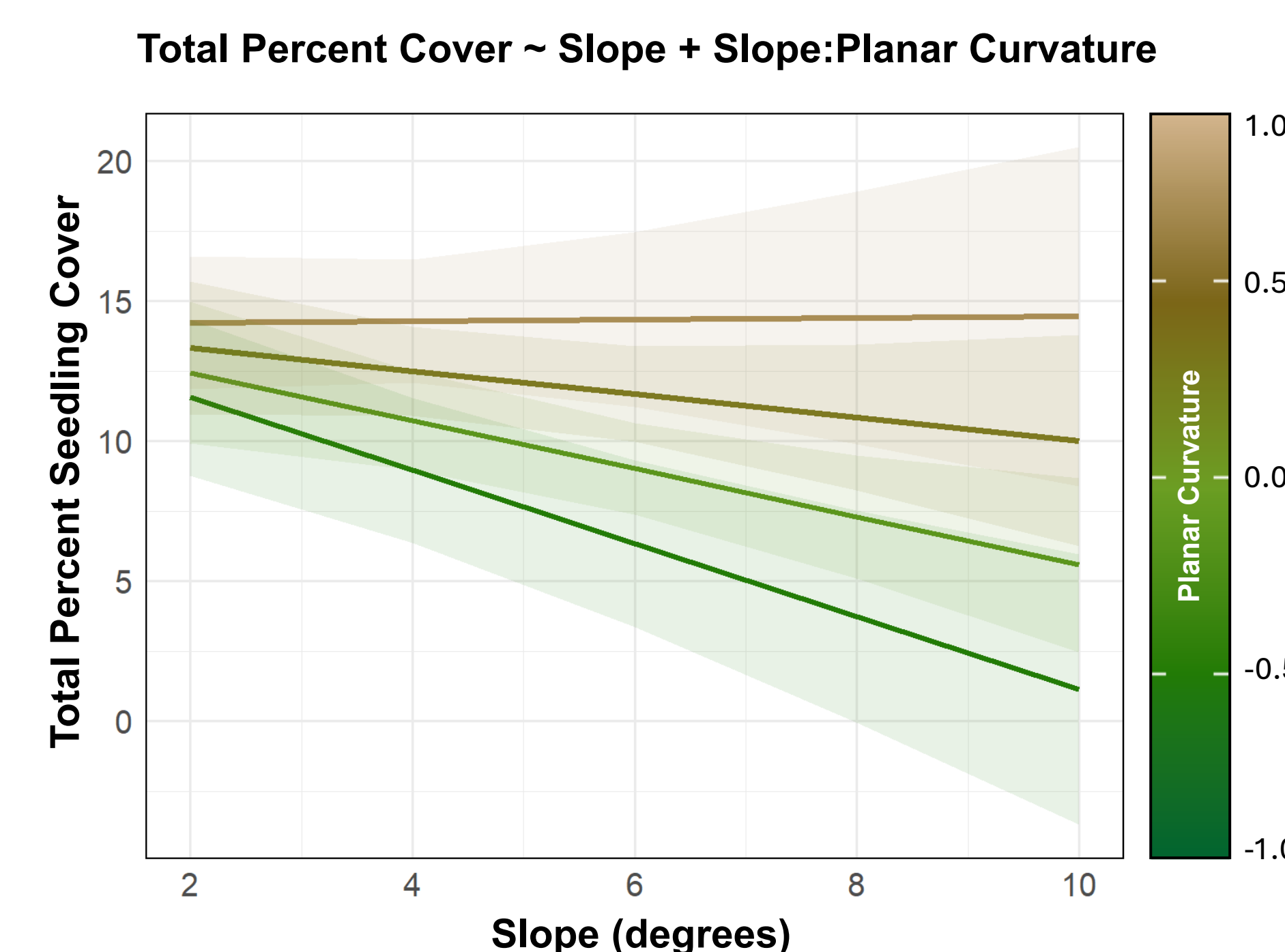


Fig. 2. The relationship between slope and total percent seedling cover depends on coinciding planar curvature.

Slope exhibited significant, negative associations with both total percent seedling cover ($p < .05$, $n = 156$) and species richness ($p < .05$, $n = 577$) (Fig. 3).

The interaction effect of slope and planar curvature on cover was significantly greater (0.91) than its effect on richness, although associations were significant and positive in each case (Fig. 3).

A plot's combination of profile curvature and topographic wetness explained significant variations in species richness ($p < .01$) (Fig. 4), though this was not the case for cover.

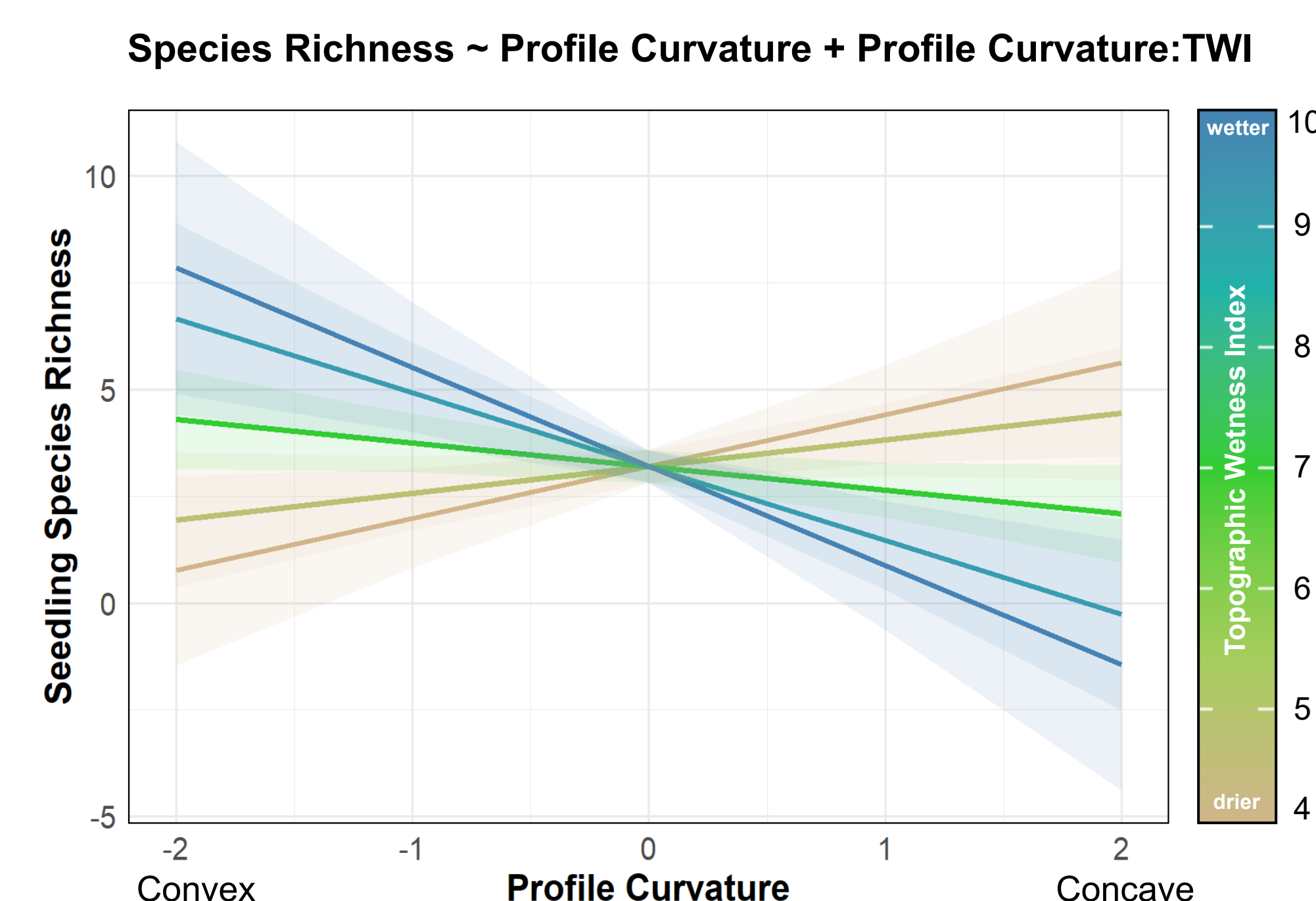


Fig. 4. The relationship between profile curvature and species richness depends on coinciding topographic wetness.

Conclusions + Future Directions



Our current results suggest that seedling species richness and cover respond differently to topographic constraints. Further, certain topographic conditions, such as curvature, modulate seedling dynamics differently depending on coinciding topographic variables, or a plot's position within the broader watershed. This work has implications for understanding how microtopography interacts with macrotopography to shape seedling performance across the landscape, and how different microtopographic variables shape seedling communities via a variety of mechanisms.

We are excited to extend this work by incorporating seedling heights, growth rates, and survival into our analyses. We expect seedlings to exhibit size- and age-structured variation in their sensitivity to topography (Inman-Narahari et al., 2014). Additionally, species-specific survival strategies may explain a population's sensitivity or affinity to certain topographic extremes (Schwartz et al., 2020).

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